



Issue Date / Ausstellungsdatum **2025-05-14** - Certificate Type / Zertifikatstyp **EN 10204**
3.1

Product / Produkt

Heat Treatment / Wärmebehandlung

Process / Prozess: **Normalizing** - Lot / Los: NORM-2024-0823-A12 - Furnace / Ofen: NF-LINE-02 - Date / Datum: 2024-08-23

Chemical Analysis / Chemische Analyse

Heat Number / Schmelzenummer **H-10988** - Melting Process / Schmelzverfahren **EEF+LF+VD** - Casting Method / Gießverfahren **ContinuousCasting**
Casting Date / Gießdatum **2025-05-09** - Sample Location / Probenort **Ladle**

$$CEV = C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15: 0.41\%$$

Mechanical Properties / Mechanische Eigenschaften

1 / 2

Elongation after fracture								EN ISO 6892-1	✓
Gauge length 5.65#S ₀ , 3 specimens tested									
No. / Nr.	1	2	3	Average / Durchschnitt	Min	Max	Std Dev / Std.-Abw.	Spec Min / Spez. Min	
Value / Wert [%]	23	24	25	24	23	25	1	19	

Reduction of Area ¹								EN ISO 6892-1	✓
No. / Nr.	1	2	3	Average / Durchschnitt	Min	Max	Std Dev / Std.-Abw.	Spec Min / Spez. Min	
Value / Wert [%]	60	62	64	62	60	64	2	50	

Charpy V-notch Impact Energy ²								EN ISO 148-1	✓
Specimen / Probekörper: Top, T-L									
No. / Nr.	1	2	3	Average / Durchschnitt	Min	Max	Std Dev / Std.-Abw.	Spec Min / Spez. Min	
Value / Wert [J]	56	58	60	58	56	60	2	40	

Charpy V-notch Impact Energy ²								EN ISO 148-1	✓
Specimen / Probekörper: Bottom, T-L									
No. / Nr.	1	2	3	Average / Durchschnitt	Min	Max	Std Dev / Std.-Abw.	Spec Min / Spez. Min	
Value / Wert [J]	54	57	59	56.7	54	59	2.5	40	

Brinell Hardness									EN ISO 6506-1		✓
Ball: 10mm, Force: 3000kg, 5 indentations											
No. / Nr.	1	2	3	4	5	Average / Durchschnitt	Min	Max	Std Dev / Std.-Abw.	Spec Min / Spez. Min	Spec Max / Spez. Max
Value / Wert [HBW]	183	185	187	184	186	185	183	187	1.58	150	220

Vickers Hardness									EN ISO 6507-1		✓
10 kg load, 5 indentations											
No. / Nr.	1	2	3	4	5	Average / Durchschnitt	Min	Max	Std Dev / Std.-Abw.	Spec Min / Spez. Min	Spec Max / Spez. Max
Value / Wert [HV10]	192	195	198	194	196	195	192	198	2.35	160	230

Rockwell Hardness	HR	18 HRC	22 HRC	EN ISO 6508-1	✓
Elastic Modulus	E	210 GPa		EN ISO 6892-1	✓
Strain Hardening Exponent	n	0.18		ASTM E646	✓
Plastic Strain Ratio	r	1.2	1.0	EN ISO 10113	✓
0.2% Proof Strength ¹				EN ISO 6892-1	✓

No. / Nr.	1	2	3	Average / Durchschnitt	Min	Max	Std Dev / Std.-Abw.	Spec Min / Spez. Min	
Value / Wert [MPa]	428	430	432	430	428	432	2	400	

¹ 3 specimens tested ² Test temperature: -20°C

Validation / Validierung

We hereby certify that the material described above has been manufactured and tested in accordance with the requirements of EN 10204:2004 type 3.1 and the specified standards. The results comply with the requirements.

Validated by / Validiert von **Johann Weber**, Quality Inspector, Quality Assurance - 2025-05-14